Heavy Tails & Inference in DBS, OCBC and UOB

- Aryan Singh, Jaanvi Bajaj, Nikhil Balaji — MDSE DS607 group coursework

- Singapore banking pillars: tail-risk choices affect capital reserves and risk limits.

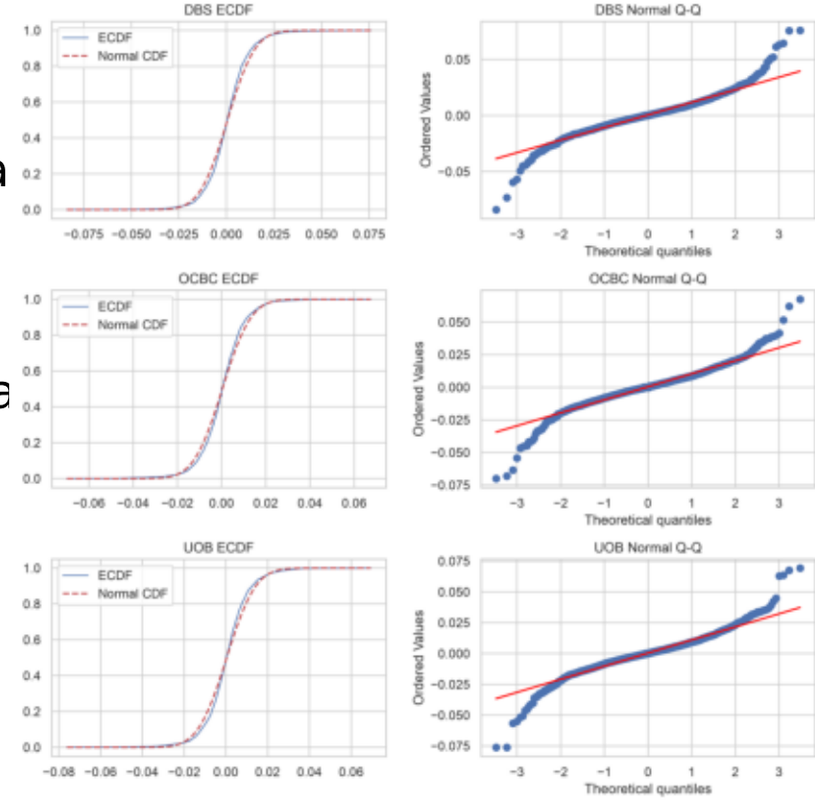
- Research question: are 2014–2024 daily returns Normal, and do CLT intervals agree with bootstrap intervals?

Repository, reproducibility & definitions

- Yahoo Finance Adjusted Close; aligned SGX dates; raw files retained offline.
- Fixed: alpha=0.05, B=10,000, 10,000 permutations, seed 42, 252 annualisation.
- Calm 2017–2019; COVID 2020-02-01 to 2020-06-30; rate-hike 2022–2023.
- Extreme dates are annotated in the cleaning log and extreme observations are retained.

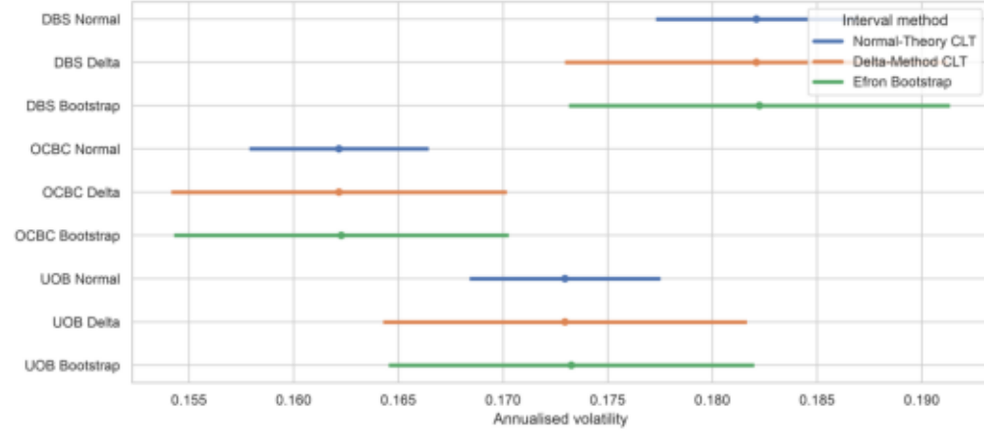
Exploratory diagnostics: heavy tails

- Daily return series exhibit volatility clustering and COVID/rate-hike regime shifts.
- Excess kurtosis is approximately 5; extreme minima to -7.4σ).
- Normal Q-Q and lower-tail log-CDF plots reveal depa



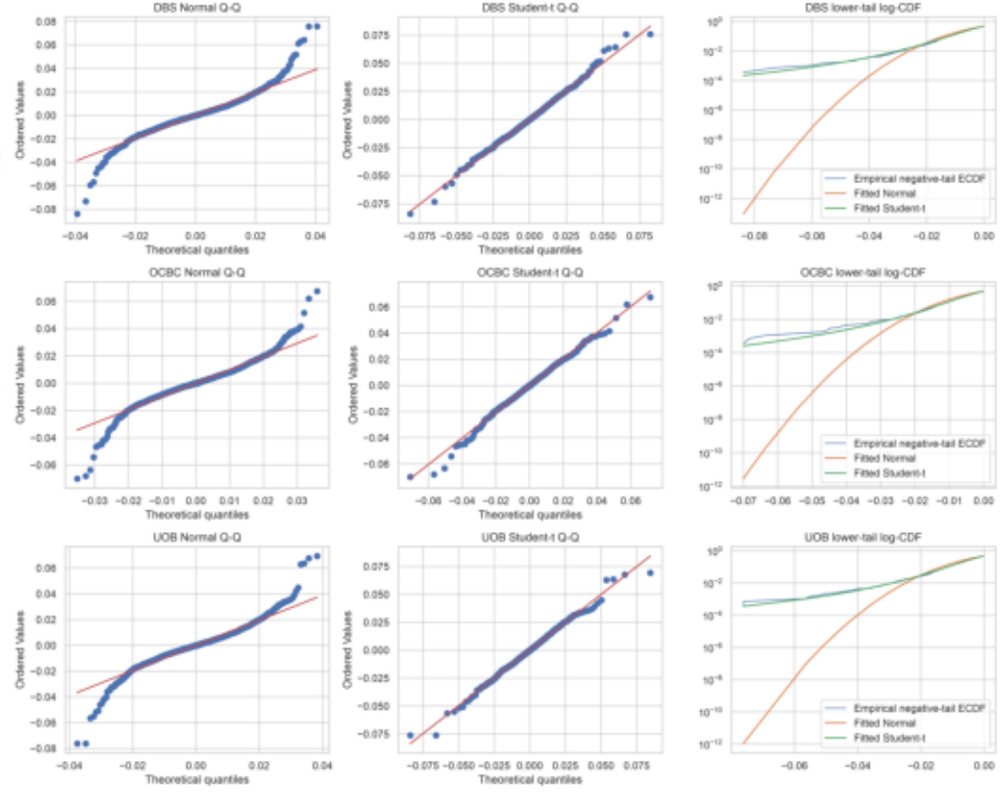
CLT vs bootstrap intervals

- Mean CLT and bootstrap intervals agree closely.
- Normal-theory volatility intervals ignore excess kurtosis and understate uncertainty.
- Under Student-t nu=5: Delta coverage is 92.5%, y 70.1%.
- Q01 is reported as an empirical 1% quantile, not a complete VaR model.



Normal vs Student-t estimation & fit

- MM nu≈5.1; MLE nu≈4.1. Fourth moments are noisy near nu=4.
- Parametric-bootstrap LRT p < 0.001 for all banks
- Student-t KS: not rejected for DBS (p≈0.066); re



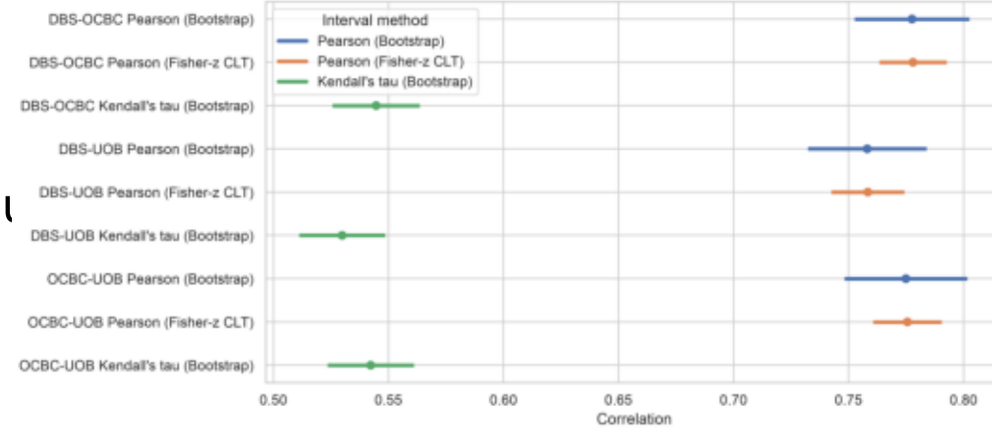
Co-movement and regime shifts

- Pearson Fisher-z CLT ≈0.03 wide; Pearson bootstrap ≈0.05 wide; Kendall measures rank association.

- COVID dispersion ratios ≈2.22–2.23; permutation p<0.0001

- OCBC rate-hike ratio 0.86; p=0.0143 does not su

- Worst-10% chi-square rejects independence for all pairs.



Takeaways and limitations

- Normal risk numbers severely understate tail risk; Student-t is substantially better but imperfect.

- Volatility clustering and clustered extreme dates undermine independence assumptions.

- One market and 11-year period; multiple tests across pairs and regimes require multiplicity caution.

Appendix: Bayesian extreme co-crashes

- Three unordered pairs: DBS–OCBC, DBS–UOB, OCBC–UOB.
- Flat Beta-Binomial agrees closely with Wald and
- Skeptical Beta(2,18) illustrates prior sensitivity.

